1 (a)	$P_0 = 300$	B1 (1)
(b)	$420 = \dfrac{900e^{0.12t}}{2e^{0.12t}+1} \Rightarrow 60e^{0.12t} = 420$	M1 A1
	Correct use of lns $\Rightarrow t = \dfrac{\ln 7}{0.12} = 16.22$	dM1 A1 (4)
(c)	States that maximum number (upper limit) is 450 so cannot reach 500	B1 (1)
		6 marks

2.(a)	$\mathrm{fg}\left(\mathrm{e}^2\right)=\mathrm{f}\left(\frac{5}{2}\ln \mathrm{e}^2\right)=\frac{12}{\frac{5}{2}\ln \mathrm{e}^2+1},=2$	M1, A1 (2)
(b)	$\mathrm{f}(x)=\frac{12}{x+1}$ $\mathrm{f}^{-1}(x)=\frac{12}{x}-1$ $0<x<12$	M1 A1 B1 (3)
(c)	$\frac{12}{x+1}=\frac{12}{x}-1\Rightarrow 12x=12(x+1)-x(x+1)$ $\Rightarrow x^2+x-12=0\Rightarrow x=\ldots$ Must be 3TQ $x=3$ only	M1 dM1 A1 (3) **8 marks**
(c) Alts	Solves $\mathrm{f}^{-1}(x)=x\Rightarrow\frac{12}{x}-1=x$ leading to quadratic equation, or solves $\mathrm{f}(x)=x\Rightarrow\frac{12}{x+1}=x$ leading to quadratic equation $\Rightarrow x^2+x-12=0\Rightarrow x=\ldots$ Must be 3TQ $x=3$ only	M1 dM1 A1 **(3)**

3.(a)	Implies equation of line is of the form $\log_{10} y = \pm\frac{2}{3}\log_{10} x + 4$	M1
	States $\log_{10} y = -\frac{2}{3}\log_{10} x + 4$ o.e.	A1
		(2)
(b)	Applies one correct log law E.g $\log_{10} y = -\frac{2}{3}\log_{10} x + 4 \rightarrow \log_{10} y = \log_{10} x^{-\frac{2}{3}} + 4$	M1
	Full attempt to undo the logs $\log_{10} y = \log_{10} x^{-\frac{2}{3}} + \log_{10} 10^4 \rightarrow y = x^{-\frac{2}{3}} \times 10^4$	dM1
	$\rightarrow y = 10\,000x^{-\frac{2}{3}}$ o.e.	A1
		(3)
		5 marks

4. (i) (a)	$f'(x)=\dfrac{4(x-3)(2x+5)-(2x+5)^2}{(x-3)^2}$ or $\dfrac{(x-3)(8x+20)-(4x^2+20x+25)}{(x-3)^2}$	M1 A1
	$=\dfrac{(2x+5)(2x-17)}{(x-3)^2}$	M1 A1
(b)	Attempts both critical values or finds one "correct" end	M1
	$x<-2.5,\ x>8.5$ (accept $x\leqslant -2.5,\ x\geqslant 8.5$)	A1
		(6)
(ii)	Attempts the chain rule on $(\sin 4x)^{\frac{1}{2}} \rightarrow A(\sin 4x)^{-\frac{1}{2}}\times\cos 4x$	M1
	$g(x)=x(\sin 4x)^{\frac{1}{2}} \Rightarrow g'(x)=(\sin 4x)^{\frac{1}{2}}+x\times\dfrac{1}{2}(\sin 4x)^{-\frac{1}{2}}\,4\cos 4x$	M1 A1
	Sets $g'(x)=0 \rightarrow (\sin 4x)^{\frac{1}{2}}+x\times\dfrac{2\cos 4x}{(\sin 4x)^{\frac{1}{2}}}=0$ and $\times\dfrac{(\sin 4x)^{\frac{1}{2}}}{\cos 4x}$ oe	M1
	$\rightarrow \tan 4x+2x=0$	A1
		(5)
		11 marks

5 (a)	$12\tan 2x + 5\cot x \sec^2 x = 0$ $12 \times \dfrac{2t}{1-t^2} + 5 \times \dfrac{1}{t}\left(1+t^2\right) = 0$ $12 \times 2t^2 + 5\left(1+t^2\right)\left(1-t^2\right) = 0 \rightarrow 5t^4 - 24t^2 - 5 = 0$ *	B1 M1 A1 A1* **(4)**
(b)	$5t^4 - 24t^2 - 5 = 0$ $\left(5t^2+1\right)\left(t^2-5\right) = 0$ Correct order of operations $t = (\pm)\sqrt{5} \Rightarrow x = ..$ Two of awrt $x = 66°, 114°, 246°, 294°$ All four of awrt $x = 65.9°, 114.1°, 245.9°, 294.1°$	M1 dM1 A1 A1 **(4)** **8 marks**

6.(a)	$(2.5, 3)$ oe	B1 B1 (2)
(b)	Attempts one solution usually $4x-10+3=3x-2 \Rightarrow x=5$	M1 A1
	Attempts both solutions $-4x+10+3=3x-2 \Rightarrow x=\frac{15}{7}$	dM1 A1 (4)
(c)	Attempts to solve $y=kx+2$ with $x=2.5, y=3$ or states that $k<4$	M1
	$k...\frac{2}{5}$	A1
	States $\frac{2}{5}<k<4$	A1 (3)
		9 marks

7. (a)	$y\big\vert_{0.8} = 2\cos 2.4 - 2.4 + 4 = 0.13$ AND $y\big\vert_{0.9} = 2\cos 2.7 - 2.7 + 4 = -0.51$	M1
	States change of sign, continuous and hence root	A1
		(2)
(b) (i)	$\left(x_2\right) = \frac{1}{3}\arccos\left(1.5\times 0.8 - 2\right) = 0.8327$	M1 A1
(ii)	$x_5 = 0.8110$	A1
		(3)
(c)	$\frac{\mathrm{d}y}{\mathrm{d}x} = -6\sin 3x - 3$	M1 A1
	Attempts $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Rightarrow \sin 3x = -\frac{1}{2} \Rightarrow x = ...$ via correct order of operations	dM1
	Achieves either $\frac{7\pi}{18}$ or $\frac{19\pi}{18}$	A1
	Correct attempt to find the third positive solution e.g. $x = \frac{\frac{\pi}{6} + 3\pi}{3}$	ddM1
	$\beta = \frac{7\pi}{18}$ and $\lambda = \frac{19\pi}{18}$	A1
		(6)
		11 marks

8.(i)	$\int \frac{2}{3x-1}\,\mathrm{d}x = \frac{2}{3}\ln(3x-1)$	M1 A1
	$\int_3^{42} \frac{2}{3x-1}\,\mathrm{d}x = \frac{2}{3}\ln(125) - \frac{2}{3}\ln(8) = \ln\frac{25}{4}$	dM1 A1
		(4)
(ii)	$\frac{2x^3-7x^2+8x+1}{(x-1)^2} = 2x+B+\frac{C}{(x-1)^2}$	B1
	Full method to find values of A, B and C	M1
	$\frac{2x^3-7x^2+8x+1}{(x-1)^2} = 2x-3+\frac{4}{(x-1)^2}$	A1
	$\int \mathrm{h}(x)\,\mathrm{d}x = \int Ax+B+\frac{C}{(x-1)^2}\,\mathrm{d}x$	
	$= \frac{1}{2}Ax^2+Bx-\frac{C}{(x-1)}$	M1 A1 ft
	$= x^2-3x-\frac{4}{(x-1)} \quad (+c)$	A1
		(6)
		10 marks
(ii) Alt I first 3 marks	$2x^3-7x^2+8x+1 = Ax(x-1)^2+B(x-1)^2+C$	
	Any of $A=2, B=-3$, or $C=4$	B1
	Either substitution or equating coefficients to get two values	M1
	All values correct $A=2, B=-3, C=4$	A1
		(3)
(ii) Alt II first 3 marks	$x^2-2x+1 \overline{)\ 2x^3-7x^2+8x+1}$ $\quad 2x-3$ $\underline{\qquad\qquad}$ 4	
	Likely to be $2x$	B1
	For attempt at division	M1
	Correct quotient and remainder	A1
		(3)

9.(a)	$R = \sqrt{41}$	B1
	$\tan\alpha = \frac{4}{5} \Rightarrow \alpha =$ awrt 0.675	M1A1
		(3)
(b)	(i) Describes stretch: stretch in the y direction by "$\sqrt{41}$"	B1 ft
	(ii) Describes translation: E.g. translate by $\begin{pmatrix} -\arctan\frac{4}{5} \\ 0 \end{pmatrix}$	B1 ft
		(2)
(c)	Attempts either $g(\theta) = \frac{90}{4 + \left(\sqrt{41}\right)^2}$ OR $g(\theta) = \frac{90}{4}$	M1
	Range $2 \leqslant g(\theta) \leqslant 22.5$	A1
		(2)
		7 marks